BEAMLINE

BUILD DOCUMENT · 25 OCTOBER 2026 DESIGNED IN BEAMLINE

A build manual for a tabletop laser-optics setup

sr ion apparatus

Overview

Sr-88+ trapped-ion laser system (after the PyOpticL sr88_apparatus): 422 nm cooling/detection, 422 nm Raman qubit beams, 461 + 405 nm photoionization, 1092 nm repump, 1033 nm quench and 674 nm SPAM, each a fiber-in/fiber-out module on a shared 1200 x 900 mm table.

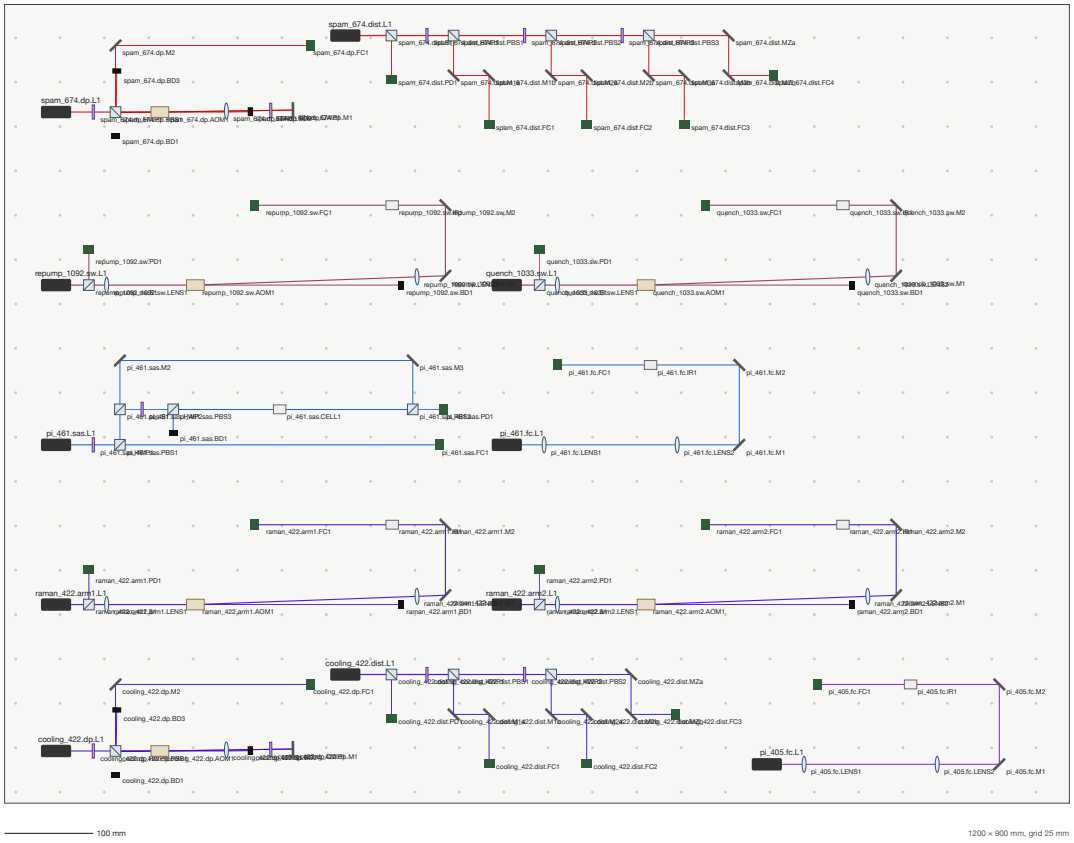


Figure 1: Plan view. Beams coloured by wavelength (infrared shown dark magenta), width ∝ log power; dashed = leaves the plate; grey dots = table holes.

Key specifications

Quantity	Value
Wavelengths	405 nm, 421.7 nm, 460.9 nm, 674 nm, 1033 nm, 1091.8 nm
Input power	280 mW from 11 source(s)
Delivered (fibers + detectors)	237 mW
Elements / beam segments	121 / 134
Baseplate	1200 × 900 × 25.4 mm, 30 printed tile(s)
Fiber cooling_422.dp.FC1	12.5 mW, coupling 98 %
Fiber cooling_422.dist.FC1	12.6 mW, coupling 99 %
Fiber cooling_422.dist.FC2	12.5 mW, coupling 98 %
Fiber cooling_422.dist.FC3	12.4 mW, coupling 98 %
Fiber pi_405.fc.FC1	9.9 mW, coupling 99 %
Fiber raman_422.arm1.FC1	22.8 mW, coupling 100 %
Fiber raman_422.arm2.FC1	22.8 mW, coupling 100 %
Fiber pi_461.sas.FC1	18 mW, coupling 100 %
Fiber pi_461.fc.FC1	9.87 mW, coupling 99 %
Fiber repump_1092.sw.FC1	22.7 mW, coupling 100 %
Fiber quench_1033.sw.FC1	22.7 mW, coupling 100 %
Fiber spam_674.dp.FC1	12 mW, coupling 94 %
Fiber spam_674.dist.FC1	9.33 mW, coupling 98 %

Fiber spam_674.dist.FC2	9.13 mW, coupling 96 %
Fiber spam_674.dist.FC3	8.88 mW, coupling 93 %
Fiber spam_674.dist.FC4	8.76 mW, coupling 92 %

Verification

PASS score 99 / 100 (Beamline tracer: 2D ray + Gaussian beam + Jones polarization)

Targets

Target	Status
met: cooling_422.dp.FC1: 12.490 mW (need ≥ 8.000)	met
met: cooling_422.dp.FC1: coupling 97.6% (need ≥ 80.0%)	met
met: cooling_422.dp.FC1: polarization V (need V, overlap 100.0%)	met
met: cooling_422.dist.FC1: 12.575 mW (need ≥ 8.870)	met
met: cooling_422.dist.FC2: 12.471 mW (need ≥ 8.870)	met
met: cooling_422.dist.FC3: 12.418 mW (need ≥ 8.870)	met
met: pi_405.fc.FC1: coupling 99.0% (need ≥ 80.0%)	met
met: pi_405.fc.FC1: 9.902 mW (need ≥ 8.000)	met
met: raman_422.arm1.PD1: 1.500 mW (need ≥ 1.000)	met
met: raman_422.arm1.FC1: 22.755 mW (need ≥ 15.000)	met
met: raman_422.arm1.FC1: coupling 99.8% (need ≥ 80.0%)	met
met: raman_422.arm2.PD1: 1.500 mW (need ≥ 1.000)	met
met: raman_422.arm2.FC1: 22.755 mW (need ≥ 15.000)	met
met: raman_422.arm2.FC1: coupling 99.8% (need ≥ 80.0%)	met
met: pi_461.sas.PD1: 0.120 mW (need ≥ 0.100)	met
met: pi_461.sas.FC1: 17.976 mW (need ≥ 10.000)	met
met: pi_461.fc.FC1: coupling 98.7% (need ≥ 80.0%)	met
met: pi_461.fc.FC1: 9.873 mW (need ≥ 8.000)	met
met: repump_1092.sw.PD1: 1.500 mW (need ≥ 1.000)	met
met: repump_1092.sw.FC1: 22.704 mW (need ≥ 15.000)	met
met: repump_1092.sw.FC1: coupling 99.6% (need ≥ 80.0%)	met
met: quench_1033.sw.PD1: 1.500 mW (need ≥ 1.000)	met
met: quench_1033.sw.FC1: 22.710 mW (need ≥ 15.000)	met
met: quench_1033.sw.FC1: coupling 99.6% (need ≥ 80.0%)	met
met: spam_674.dp.FC1: 12.035 mW (need ≥ 8.000)	met
met: spam_674.dp.FC1: coupling 94.0% (need ≥ 80.0%)	met
met: spam_674.dp.FC1: polarization V (need V, overlap 100.0%)	met
met: spam_674.dist.FC1: 9.326 mW (need ≥ 6.650)	met
met: spam_674.dist.FC2: 9.133 mW (need ≥ 6.650)	met
met: spam_674.dist.FC3: 8.880 mW (need ≥ 6.650)	met
met: spam_674.dist.FC4: 8.762 mW (need ≥ 6.650)	met

Checks

Severity	Check	Message
INFO	target	met: cooling_422.dp.FC1: 12.490 mW (need ≥ 8.000)
INFO	target	met: cooling_422.dp.FC1: coupling 97.6% (need ≥ 80.0%)
INFO	target	met: cooling_422.dp.FC1: polarization V (need V, overlap 100.0%)
INFO	target	met: cooling_422.dist.FC1: 12.575 mW (need ≥ 8.870)
INFO	target	met: cooling_422.dist.FC2: 12.471 mW (need ≥ 8.870)
INFO	target	met: cooling_422.dist.FC3: 12.418 mW (need ≥ 8.870)
INFO	target	met: pi_405.fc.FC1: coupling 99.0% (need ≥ 80.0%)
INFO	target	met: pi_405.fc.FC1: 9.902 mW (need ≥ 8.000)
INFO	target	met: raman_422.arm1.PD1: 1.500 mW (need ≥ 1.000)
INFO	target	met: raman_422.arm1.FC1: 22.755 mW (need ≥ 15.000)
INFO	target	met: raman_422.arm1.FC1: coupling 99.8% (need ≥ 80.0%)
INFO	target	met: raman_422.arm2.PD1: 1.500 mW (need ≥ 1.000)
INFO	target	met: raman_422.arm2.FC1: 22.755 mW (need ≥ 15.000)
INFO	target	met: raman_422.arm2.FC1: coupling 99.8% (need ≥ 80.0%)
INFO	target	met: pi_461.sas.PD1: 0.120 mW (need ≥ 0.100)
INFO	target	met: pi_461.sas.FC1: 17.976 mW (need ≥ 10.000)
INFO	target	met: pi_461.fc.FC1: coupling 98.7% (need ≥ 80.0%)
INFO	target	met: pi_461.fc.FC1: 9.873 mW (need ≥ 8.000)

INFO	target	met: repump_1092.sw.PD1: 1.500 mW (need ≥ 1.000)
INFO	target	met: repump_1092.sw.FC1: 22.704 mW (need ≥ 15.000)
INFO	target	met: repump_1092.sw.FC1: coupling 99.6% (need ≥ 80.0%)
INFO	target	met: quench_1033.sw.PD1: 1.500 mW (need ≥ 1.000)
INFO	target	met: quench_1033.sw.FC1: 22.710 mW (need ≥ 15.000)
INFO	target	met: quench_1033.sw.FC1: coupling 99.6% (need ≥ 80.0%)
INFO	target	met: spam_674.dp.FC1: 12.035 mW (need ≥ 8.000)
INFO	target	met: spam_674.dp.FC1: coupling 94.0% (need ≥ 80.0%)
INFO	target	met: spam_674.dp.FC1: polarization V (need V, overlap 100.0%)
INFO	target	met: spam_674.dist.FC1: 9.326 mW (need ≥ 6.650)
INFO	target	met: spam_674.dist.FC2: 9.133 mW (need ≥ 6.650)
INFO	target	met: spam_674.dist.FC3: 8.880 mW (need ≥ 6.650)
INFO	target	met: spam_674.dist.FC4: 8.762 mW (need ≥ 6.650)

Bill of materials

Purchased parts

Part	Description	Vendor	Qty	Unit	Line
AC127-050-B	lens (Telescope lens f=50 mm)	Thorlabs	2	$90.00	$180.00
AC127-075-B	lens (Cat-eye lens f=75 mm)	Thorlabs	2	$90.00	$180.00
AC127-100-B	lens (Telescope lens f=100 mm)	Thorlabs	6	$90.00	$540.00
AC254-250-B	lens (Recollimating lens f=250 mm)	Thorlabs	4	$110.00	$440.00
BB05-E02	mirror (Tap 1 steering 1)	Thorlabs	14	$65.00	$910.00
BB05-E03	mirror (Cat-eye retro mirror (focal plane))	Thorlabs	18	$65.00	$1,170.00
BSF05-A	beam sampler (5% pickoff)	Thorlabs	2	$55.00	$110.00
BSF05-B	beam sampler (5% pickoff)	Thorlabs	5	$55.00	$275.00
GC25075-RB	vapor cell (Rb vapor cell, 75 mm)	Thorlabs	1	$750.00	$750.00
ID12	Ø12 mm mounted iris diaphragm	Thorlabs	6	$75.00	$450.00
KM100	kinematic mirror mount, Ø1"	Thorlabs	39	$48.00	$1,872.00
KM100CP	kinematic mount for 10 mm cubes	Thorlabs	10	$70.00	$700.00
LB1	beam block / dump, black anodized	Thorlabs	11	$95.00	$1,045.00
LMR1	lens mount, Ø1" optics	Thorlabs	14	$18.00	$252.00
PAF2-11B	fiber coupler (Output fiber coupler (PM780-HP))	Thorlabs	5	$420.00	$2,100.00
PAF2-A4A	Fiber input (PM780-HP via PAF2-A4A collimator) (421.7 nm, 20 mW)	Thorlabs	19	$420.00	$7,980.00
PBS052	10 mm polarizing beamsplitter cube, 620-1000 nm	Thorlabs	5	$260.00	$1,300.00
PBS101	polarizing beamsplitter (Tap 1 PBS)	Thorlabs	5	$260.00	$1,300.00
PDA36A2	Si amplified switchable-gain photodetector	Thorlabs	7	$360.00	$2,520.00
RSP05	rotation mount, Ø1/2" optics	Thorlabs	11	$95.00	$1,045.00
WPH05M	half-wave plate (lambda/2 tap 1)	Thorlabs	5	$310.00	$1,550.00
WPH05M-780	half-wave plate (Input λ/2 (rotate to H))	Thorlabs	4	$310.00	$1,240.00
WPQ05M-780	quarter-wave plate (λ/4 at 45° (double pass → λ/2))	Thorlabs	2	$290.00	$580.00
Isomet 1205C-2	acousto-optic modulator (80 MHz AOM)	Isomet	2	$1,500.00	$3,000.00
MT200-A0.5-800	acousto-optic modulator (200 MHz AOM)	AA Opto-Electronic	4	$1,500.00	$6,000.00
Thorlabs LP785-SF20 + C230TMD-B	Collimated laser (405 nm, 10 mW)		2	—	—
Toptica DL pro 780	ECDL output (collimated, after isolator) (460.9 nm, 20 mW)		1	—	—

Estimated total: $37,760.50

list-price estimates, USD, excluding lasers; verify with vendors

Hardware

Part	Description	Qty	Unit
SH25S050	1/4-20 x 1/2" cap screw (25 pack -> per piece)	405	$0.40
SH8S100	8-32 x 1" cap screw (per piece)	121	$0.30
DOWEL-2x8	Ø2 x 8 mm steel dowel pin (per piece)	240	$0.15
M3x25+NUT	M3 x 25 mm cap screw + nut (tile lap joints)	186	$0.20

3D-printed parts

Part	STL file	cm³	Print h	PLA g
baseplate_r1c1	stl/baseplate_r1c1.stl	996.5	26.6	396
baseplate_r2c1	stl/baseplate_r2c1.stl	903.9	24.5	364
baseplate_r3c1	stl/baseplate_r3c1.stl	938.6	25.6	381
baseplate_r4c1	stl/baseplate_r4c1.stl	990.9	26.9	400
baseplate_r5c1	stl/baseplate_r5c1.stl	874.2	23.5	349
baseplate_r1c2	stl/baseplate_r1c2.stl	927.4	25	373
baseplate_r2c2	stl/baseplate_r2c2.stl	842.9	23.1	344
baseplate_r3c2	stl/baseplate_r3c2.stl	878	24	357
baseplate_r4c2	stl/baseplate_r4c2.stl	925.2	25.3	377
baseplate_r5c2	stl/baseplate_r5c2.stl	817.7	22.2	331
baseplate_r1c3	stl/baseplate_r1c3.stl	794.7	22	327
baseplate_r2c3	stl/baseplate_r2c3.stl	724.5	20.2	301
baseplate_r3c3	stl/baseplate_r3c3.stl	757.4	21	312
baseplate_r4c3	stl/baseplate_r4c3.stl	796	22.2	330
baseplate_r5c3	stl/baseplate_r5c3.stl	709.8	19.4	289
baseplate_r1c4	stl/baseplate_r1c4.stl	972.1	26.4	392
baseplate_r2c4	stl/baseplate_r2c4.stl	885.4	24.3	361
baseplate_r3c4	stl/baseplate_r3c4.stl	925.9	25.2	375
baseplate_r4c4	stl/baseplate_r4c4.stl	973.5	26.6	396
baseplate_r5c4	stl/baseplate_r5c4.stl	871.5	23.3	346
baseplate_r1c5	stl/baseplate_r1c5.stl	908.4	24.7	367
baseplate_r2c5	stl/baseplate_r2c5.stl	825	22.8	340
baseplate_r3c5	stl/baseplate_r3c5.stl	861.6	23.7	352
baseplate_r4c5	stl/baseplate_r4c5.stl	907	25	372
baseplate_r5c5	stl/baseplate_r5c5.stl	810.2	21.8	324
baseplate_r1c6	stl/baseplate_r1c6.stl	979.7	26.1	389
baseplate_r2c6	stl/baseplate_r2c6.stl	896.3	24.2	359
baseplate_r3c6	stl/baseplate_r3c6.stl	934.2	25.1	373
baseplate_r4c6	stl/baseplate_r4c6.stl	985.3	26.5	394
baseplate_r5c6	stl/baseplate_r5c6.stl	874.9	23.2	345

Assembly

Step 1 **Print the baseplate** — Print 30 file(s) from stl/ flat, top side up: PLA, 20 % infill, 3 walls (≈ 720.4 h, ≈ 10716 g).

Step 2 **Join the tiles** — Lay the tiles out by their r/c names, overlap the 20 mm half-lap joints and fasten with 186 M3 × 25 bolts + nuts.

Step 3 **Press in the dowel pins** — Press 240 Ø2 × 8 mm pins into the paired pin holes; they key each mount's orientation.

Step 4 **Bolt to the table** — Fix the plate with 405 1/4-20 (or M6) cap screws through the counterbored holes (2-grid pitch, half a grid in from the edges).

Step 5 **cooling_422.dp.HWP1** — Mount Input λ/2 (RSP05, WPH05M-780) at site 1, (100.0, 58.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Set the fast axis to 10.0° from horizontal (H).

Step 6 **cooling_422.dp.PBS1** — Mount Polarizing beamsplitter cube (KM100CP, PBS052) at site 2, (125.0, 58.8) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Outputs: cooling_422.dp.L1.t → 0.0° (20 mW); cooling_422.dp.L1.t.+1.0.r → 89.1° (3.2 mW); cooling_422.dp.L1.t.+1.+1.r → 90.0° (12.8 mW)

Step 7 **cooling_422.dp.AOM1** — Mount 80 MHz AOM (Isomet 1205C-2) at site 3, (175.0, 58.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Drive at 148.02 MHz; use the +1 order (≈ 80 % efficiency).

Step 8 **cooling_422.dp.LENS1** — Mount Cat-eye lens f=75 mm (AC127-075-B, LMR1) at site 4, (250.0, 59.9) mm; orientation 0.9°; keyed by pins, one 8-32 screw from below. f = 75 mm, 75.0 mm after the previous element on cooling_422.dp.L1.t.+1.

Step 9 **cooling_422.dp.BD2** — Mount 0th-order razor block (LB1) at site 5, (275.0, 59.2) mm; orientation 0.9°; keyed by pins, one 8-32 screw from below. Blocks: cooling_422.dp.L1.t.0 (4 mW, 421.7 nm)

Step 10 **cooling_422.dp.QWP1** — Mount λ/4 at 45° (RSP05, WPQ05M-780) at site 6, (300.0, 60.7) mm; orientation 0.9°; keyed by pins, one 8-32 screw from below. Set the fast axis to 45.0° from horizontal (H).

Step 11 **cooling_422.dp.M1** — Mount Cat-eye retro mirror (BB05-E03, KM100) at site 7, (325.0, 61.0) mm; orientation 180.9°; keyed by pins, one 8-32 screw from below. Steer the beam to: cooling_422.dp.L1.t.+1 → 180.9° (16 mW)

Step 12 **cooling_422.dp.BD3** — Mount Return 0th-order ghost block (LB1) at site 8, (126.4, 103.1) mm; orientation 89.1°; keyed by pins, one 8-32 screw from below. Blocks: cooling_422.dp.L1.t.+1.0.r (3.2 mW, 421.7 nm)

Step 13 **cooling_422.dp.M2** — Mount Output fold mirror (BB05-E03, KM100) at site 9, (125.0, 133.8) mm; orientation 315.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: cooling_422.dp.L1.t.+1.+1.r → 360.0° (12.8 mW)

Step 14 **cooling_422.dp.FC1** — Mount Output fiber coupler (PAF2-A4A) at site 10, (340.0, 133.8) mm; orientation 360.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 4.6 mm, fiber MFD 2.86 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 12.5 mW (98 % coupling).

Step 15 **cooling_422.dist.S1** — Mount 5% pickoff (BSF05-A, KM100) at site 11, (436.0, 145.0) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Outputs: cooling_422.dist.L1.t → 0.0° (38 mW); cooling_422.dist.L1.r → 270.0° (2 mW)

Step 16 **cooling_422.dist.HWP1** — Mount lambda/2 tap 1 (RSP05, WPH05M) at site 12, (476.0, 145.0) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Set the fast axis to 17.6° from horizontal (H).

Step 17 **cooling_422.dist.PBS1** — Mount Tap 1 PBS (KM100CP, PBS101) at site 13, (506.0, 145.0) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Outputs: cooling_422.dist.L1.t.t → 0.0° (25.3 mW); cooling_422.dist.L1.t.r → 270.0° (12.7 mW)

Step 18 **cooling_422.dist.HWP2** — Mount lambda/2 tap 2 (RSP05, WPH05M) at site 14, (586.0, 145.0) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Set the fast axis to 22.5° from horizontal (H).

Step 19 **cooling_422.dist.PBS2** — Mount Tap 2 PBS (KM100CP, PBS101) at site 15, (616.0, 145.0) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Outputs: cooling_422.dist.L1.t.t.t → 0.0° (12.7 mW); cooling_422.dist.L1.t.t.r → 270.0° (12.7 mW)

Step 20 **cooling_422.dist.MZa** — Mount Last output steering 1 (BB05-E02, KM100) at site 16, (706.0, 145.0) mm; orientation 225.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: cooling_422.dist.L1.t.t.t → 270.0° (12.7 mW)

Step 21 **cooling_422.dist.MZb** — Mount Last output steering 2 (BB05-E02, KM100) at site 17, (706.0, 100.0) mm; orientation 45.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: cooling_422.dist.L1.t.t.t → 0.0° (12.7 mW)

Step 22 **cooling_422.dist.FC3** — Mount Output fiber 3 (PAF2-A4A) at site 18, (751.0, 100.0) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 4.6 mm, fiber MFD 2.87 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 12.4 mW (98 % coupling).

Step 23 **cooling_422.dist.M2a** — Mount Tap 2 steering 1 (BB05-E02, KM100) at site 19, (616.0, 100.0) mm; orientation 45.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: cooling_422.dist.L1.t.t.r → 0.0° (12.7 mW)

Step 24 **cooling_422.dist.M2b** — Mount Tap 2 steering 2 (BB05-E02, KM100) at site 20, (656.0, 100.0) mm; orientation 225.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: cooling_422.dist.L1.t.t.r → 270.0° (12.7 mW)

Step 25 **cooling_422.dist.FC2** — Mount Output fiber 2 (PAF2-A4A) at site 21, (656.0, 50.0) mm; orientation 270.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 4.6 mm, fiber MFD 2.87 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 12.5 mW (98 % coupling).

Step 26 **cooling_422.dist.M1a** — Mount Tap 1 steering 1 (BB05-E02, KM100) at site 22, (506.0, 100.0) mm; orientation 45.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: cooling_422.dist.L1.t.r → 0.0° (12.7 mW)

Step 27 **cooling_422.dist.M1b** — Mount Tap 1 steering 2 (BB05-E02, KM100) at site 23, (546.0, 100.0) mm; orientation 225.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: cooling_422.dist.L1.t.r → 270.0° (12.7 mW)

Step 28 **cooling_422.dist.FC1** — Mount Output fiber 1 (PAF2-A4A) at site 24, (546.0, 50.0) mm; orientation 270.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 4.6 mm, fiber MFD 2.87 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 12.6 mW (99 % coupling).

Step 29 **cooling_422.dist.PD1** — Mount Power monitor (PDA36A2) at site 25, (436.0, 100.0) mm; orientation 270.0°; keyed by pins, one 8-32 screw from below. Expect ≈ 2 mW on the detector.

Step 30 **pi_405.fc.LENS1** — Mount Telescope lens f=100 mm (AC127-100-B, LMR1) at site 26, (901.0, 43.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. f = 100 mm, 25.0 mm after the previous element on pi_405.fc.L1.

Step 31 **pi_405.fc.LENS2** — Mount Telescope lens f=50 mm (AC127-050-B, LMR1) at site 27, (1051.0, 43.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. f = 50 mm, 150.0 mm after the previous element on pi_405.fc.L1.

Step 32 **pi_405.fc.M1** — Mount Steering mirror 1 (BB05-E03, KM100) at site 28, (1121.0, 43.8) mm; orientation 135.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: pi_405.fc.L1 → 90.0° (10 mW)

Step 33 **pi_405.fc.M2** — Mount Steering mirror 2 (BB05-E03, KM100) at site 29, (1121.0, 133.8) mm; orientation 225.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: pi_405.fc.L1 → 180.0° (10 mW)

Step 34 **pi_405.fc.IR1** — Mount Alignment iris (ID12) at site 30, (1021.0, 133.8) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below.

Step 35 **pi_405.fc.FC1** — Mount Fiberport (PAF2-A4A) at site 31, (921.0, 133.8) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 4.6 mm, fiber MFD 2.75 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 9.9 mW (99 % coupling).

Step 36 **raman_422.arm1.S1** — Mount 5% pickoff (BSF05-B, KM100) at site 32, (95.0, 223.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Outputs: raman_422.arm1.L1.t → 0.0° (28.5 mW); raman_422.arm1.L1.r → 90.0° (1.5 mW)

Step 37 **raman_422.arm1.LENS1** — Mount Focusing lens f=100 mm (AC127-100-B, LMR1) at site 33, (115.0, 223.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. f = 100 mm, 20.0 mm after the previous element on raman_422.arm1.L1.t.

Step 38 **raman_422.arm1.AOM1** — Mount 200 MHz AOM (MT200-A0.5-800) at site 34, (215.0, 223.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Drive at 370.05 MHz; use the +1 order (≈ 80 % efficiency).

Step 39 **raman_422.arm1.BD1** — Mount 0th-order dump (LB1) at site 35, (445.0, 223.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Blocks: raman_422.arm1.L1.t.0 (5.7 mW, 421.7 nm)

Step 40 **raman_422.arm1.LENS2** — Mount Recollimating lens f=250 mm (AC254-250-B, LMR1) at site 36, (464.8, 233.1) mm; orientation 2.1°; keyed by pins, one 8-32 screw from below. f = 250 mm, 250.0 mm after the previous element on raman_422.arm1.L1.t.+1.

Step 41 **raman_422.arm1.M1** — Mount +1 order pickoff mirror (BB05-E03, KM100) at site 37, (496.8, 234.3) mm; orientation 136.1°; keyed by pins, one 8-32 screw from below. Steer the beam to: raman_422.arm1.L1.t.+1 → 90.0° (22.8 mW)

Step 42 **raman_422.arm1.M2** — Mount Fiber steering mirror (BB05-E03, KM100) at site 38, (496.8, 313.8) mm; orientation 225.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: raman_422.arm1.L1.t.+1 → 180.0° (22.8 mW)

Step 43 **raman_422.arm1.IR1** — Mount Alignment iris (ID12) at site 39, (436.8, 313.8) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below.

Step 44 **raman_422.arm1.FC1** — Mount Output fiber coupler (PAF2-11B) at site 40, (286.8, 313.8) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 11 mm, fiber MFD 2.86 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 22.8 mW (100 % coupling).

Step 45 **raman_422.arm1.PD1** — Mount Intensity monitor (PDA36A2) at site 41, (95.0, 258.8) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Expect ≈ 1.5 mW on the detector.

Step 46 **raman_422.arm2.S1** — Mount 5% pickoff (BSF05-B, KM100) at site 42, (603.0, 223.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Outputs: raman_422.arm2.L1.t → 0.0° (28.5 mW); raman_422.arm2.L1.r → 90.0° (1.5 mW)

Step 47 **raman_422.arm2.LENS1** — Mount Focusing lens f=100 mm (AC127-100-B, LMR1) at site 43, (623.0, 223.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. f = 100 mm, 20.0 mm after the previous element on raman_422.arm2.L1.t.

Step 48 **raman_422.arm2.AOM1** — Mount 200 MHz AOM (MT200-A0.5-800) at site 44, (723.0, 223.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Drive at 370.05 MHz; use the +1 order (≈ 80 % efficiency).

Step 49 **raman_422.arm2.BD1** — Mount 0th-order dump (LB1) at site 45, (953.0, 223.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Blocks: raman_422.arm2.L1.t.0 (5.7 mW, 421.7 nm)

Step 50 **raman_422.arm2.LENS2** — Mount Recollimating lens f=250 mm (AC254-250-B, LMR1) at site 46, (972.8, 233.1) mm; orientation 2.1°; keyed by pins, one 8-32 screw from below. f = 250 mm, 250.0 mm after the previous element on raman_422.arm2.L1.t.+1.

Step 51 **raman_422.arm2.M1** — Mount +1 order pickoff mirror (BB05-E03, KM100) at site 47, (1004.8, 234.3) mm; orientation 136.1°; keyed by pins, one 8-32 screw from below. Steer the beam to: raman_422.arm2.L1.t.+1 → 90.0° (22.8 mW)

Step 52 **raman_422.arm2.M2** — Mount Fiber steering mirror (BB05-E03, KM100) at site 48, (1004.8, 313.8) mm; orientation 225.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: raman_422.arm2.L1.t.+1 → 180.0° (22.8 mW)

Step 53 **raman_422.arm2.IR1** — Mount Alignment iris (ID12) at site 49, (944.8, 313.8) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below.

Step 54 **raman_422.arm2.FC1** — Mount Output fiber coupler (PAF2-11B) at site 50, (794.8, 313.8) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 11 mm, fiber MFD 2.86 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 22.8 mW (100 % coupling).

Step 55 **raman_422.arm2.PD1** — Mount Intensity monitor (PDA36A2) at site 51, (603.0, 258.8) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Expect ≈ 1.5 mW on the detector.

Step 56 **pi_461.sas.HWP1** — Mount Split-ratio λ/2 (RSP05, WPH05M-780) at site 52, (100.0, 403.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Set the fast axis to 9.2° from horizontal (H).

Step 57 **pi_461.sas.PBS1** — Mount Spectroscopy pickoff PBS (KM100CP, PBS052) at site 53, (130.0, 403.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Outputs: pi_461.sas.L1.t → 0.0° (18 mW); pi_461.sas.L1.r → 90.0° (1.99 mW)

Step 58 **pi_461.sas.FC1** — Mount Main output fiber coupler (PAF2-11B) at site 54, (485.0, 403.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 11 mm, fiber MFD 3.13 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 18 mW (100 % coupling).

Step 59 **pi_461.sas.S1** — Mount Pump/probe sampler (BSF05-B, KM100) at site 55, (130.0, 443.8) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below. Outputs: pi_461.sas.L1.r.t → 90.0° (1.79 mW); pi_461.sas.L1.r.r → 360.0° (0.199 mW)

Step 60 **pi_461.sas.M2** — Mount Pump mirror 1 (BB05-E03, KM100) at site 56, (130.0, 498.8) mm; orientation 315.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: pi_461.sas.L1.r.t → 0.0° (1.79 mW)

Step 61 **pi_461.sas.M3** — Mount Pump mirror 2 (BB05-E03, KM100) at site 57, (460.0, 498.8) mm; orientation 225.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: pi_461.sas.L1.r.t → 270.0° (1.79 mW)

Step 62 **pi_461.sas.PBS2** — Mount Pump injection PBS (KM100CP, PBS052) at site 58, (460.0, 443.8) mm; orientation 360.0°; keyed by pins, one 8-32 screw from below. Outputs: pi_461.sas.L1.r.t.r → 180.0° (1.79 mW); pi_461.sas.L1.r.r.t.t → 360.0° (0.12 mW)

Step 63 **pi_461.sas.CELL1** — Mount Rb vapor cell, 75 mm (GC25075-RB) at site 59, (310.0, 443.8) mm; orientation 360.0°; keyed by pins, one 8-32 screw from below.

Step 64 **pi_461.sas.PBS3** — Mount Pump rejection PBS (KM100CP, PBS052) at site 60, (190.0, 443.8) mm; orientation 360.0°; keyed by pins, one 8-32 screw from below. Outputs: pi_461.sas.L1.r.t.r.r → 270.0° (1.08 mW); pi_461.sas.L1.r.r.t → 360.0° (0.199 mW)

Step 65 **pi_461.sas.BD1** — Mount Pump dump (LB1) at site 61, (190.0, 418.8) mm; orientation 270.0°; keyed by pins, one 8-32 screw from below. Blocks: pi_461.sas.L1.r.t.r.r (1.08 mW, 460.9 nm)

Step 66 **pi_461.sas.HWP2** — Mount Probe λ/2 (RSP05, WPH05M-780) at site 62, (155.0, 443.8) mm; orientation 360.0°; keyed by pins, one 8-32 screw from below. Set the fast axis to 45.0° from horizontal (H).

Step 67 **pi_461.sas.PD1** — Mount Probe photodiode (PDA36A2) at site 63, (490.0, 443.8) mm; orientation 360.0°; keyed by pins, one 8-32 screw from below. Expect ≈ 0.12 mW on the detector.

Step 68 **pi_461.fc.LENS1** — Mount Telescope lens f=100 mm (AC127-100-B, LMR1) at site 64, (608.0, 403.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. f = 100 mm, 25.0 mm after the previous element on pi_461.fc.L1.

Step 69 **pi_461.fc.LENS2** — Mount Telescope lens f=50 mm (AC127-050-B, LMR1) at site 65, (758.0, 403.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. f = 50 mm, 150.0 mm after the previous element on pi_461.fc.L1.

Step 70 **pi_461.fc.M1** — Mount Steering mirror 1 (BB05-E03, KM100) at site 66, (828.0, 403.8) mm; orientation 135.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: pi_461.fc.L1 → 90.0° (10 mW)

Step 71 **pi_461.fc.M2** — Mount Steering mirror 2 (BB05-E03, KM100) at site 67, (828.0, 493.8) mm; orientation 225.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: pi_461.fc.L1 → 180.0° (10 mW)

Step 72 **pi_461.fc.IR1** — Mount Alignment iris (ID12) at site 68, (728.0, 493.8) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below.

Step 73 **pi_461.fc.FC1** — Mount Fiberport (PAF2-A4A) at site 69, (628.0, 493.8) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 4.6 mm, fiber MFD 3.13 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 9.87 mW (99 % coupling).

Step 74 **repump_1092.sw.S1** — Mount 5% pickoff (BSF05-B, KM100) at site 70, (95.0, 583.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Outputs: repump_1092.sw.L1.t → 0.0° (28.5 mW); repump_1092.sw.L1.r → 90.0° (1.5 mW)

Step 75 **repump_1092.sw.LENS1** — Mount Focusing lens f=100 mm (AC127-100-B, LMR1) at site 71, (115.0, 583.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. f = 100 mm, 20.0 mm after the previous element on repump_1092.sw.L1.t.

Step 76 **repump_1092.sw.AOM1** — Mount 200 MHz AOM (MT200-A0.5-800) at site 72, (215.0, 583.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Drive at 142.93 MHz; use the +1 order (≈ 80 % efficiency).

Step 77 **repump_1092.sw.BD1** — Mount 0th-order dump (LB1) at site 73, (445.0, 583.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Blocks: repump_1092.sw.L1.t.0 (5.7 mW, 1091.8 nm)

Step 78 **repump_1092.sw.LENS2** — Mount Recollimating lens f=250 mm (AC254-250-B, LMR1) at site 74, (464.8, 593.1) mm; orientation 2.1°; keyed by pins, one 8-32 screw from below. f = 250 mm, 250.0 mm after the previous element on repump_1092.sw.L1.t.+1.

Step 79 **repump_1092.sw.M1** — Mount +1 order pickoff mirror (BB05-E03, KM100) at site 75, (496.8, 594.3) mm; orientation 136.1°; keyed by pins, one 8-32 screw from below. Steer the beam to: repump_1092.sw.L1.t.+1 → 90.0° (22.8 mW)

Step 80 **repump_1092.sw.M2** — Mount Fiber steering mirror (BB05-E03, KM100) at site 76, (496.8, 673.8) mm; orientation 225.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: repump_1092.sw.L1.t.+1 → 180.0° (22.8 mW)

Step 81 **repump_1092.sw.IR1** — Mount Alignment iris (ID12) at site 77, (436.8, 673.8) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below.

Step 82 **repump_1092.sw.FC1** — Mount Output fiber coupler (PAF2-11B) at site 78, (286.8, 673.8) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 11 mm, fiber MFD 7.42 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 22.7 mW (100 % coupling).

Step 83 **repump_1092.sw.PD1** — Mount Intensity monitor (PDA36A2) at site 79, (95.0, 618.8) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Expect ≈ 1.5 mW on the detector.

Step 84 **quench_1033.sw.S1** — Mount 5% pickoff (BSF05-B, KM100) at site 80, (603.0, 583.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Outputs: quench_1033.sw.L1.t → 0.0° (28.5 mW); quench_1033.sw.L1.r → 90.0° (1.5 mW)

Step 85 **quench_1033.sw.LENS1** — Mount Focusing lens f=100 mm (AC127-100-B, LMR1) at site 81, (623.0, 583.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. f = 100 mm, 20.0 mm after the previous element on quench_1033.sw.L1.t.

Step 86 **quench_1033.sw.AOM1** — Mount 200 MHz AOM (MT200-A0.5-800) at site 82, (723.0, 583.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Drive at 151.06 MHz; use the +1 order (≈ 80 % efficiency).

Step 87 **quench_1033.sw.BD1** — Mount 0th-order dump (LB1) at site 83, (953.0, 583.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Blocks: quench_1033.sw.L1.t.0 (5.7 mW, 1033 nm)

Step 88 **quench_1033.sw.LENS2** — Mount Recollimating lens f=250 mm (AC254-250-B, LMR1) at site 84, (972.8, 593.1) mm; orientation 2.1°; keyed by pins, one 8-32 screw from below. f = 250 mm, 250.0 mm after the previous element on quench_1033.sw.L1.t.+1.

Step 89 **quench_1033.sw.M1** — Mount +1 order pickoff mirror (BB05-E03, KM100) at site 85, (1004.8, 594.3) mm; orientation 136.1°; keyed by pins, one 8-32 screw from below. Steer the beam to: quench_1033.sw.L1.t.+1 → 90.0° (22.8 mW)

Step 90 **quench_1033.sw.M2** — Mount Fiber steering mirror (BB05-E03, KM100) at site 86, (1004.8, 673.8) mm; orientation 225.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: quench_1033.sw.L1.t.+1 → 180.0° (22.8 mW)

Step 91 **quench_1033.sw.IR1** — Mount Alignment iris (ID12) at site 87, (944.8, 673.8) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below.

Step 92 **quench_1033.sw.FC1** — Mount Output fiber coupler (PAF2-11B) at site 88, (794.8, 673.8) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 11 mm, fiber MFD 7.02 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 22.7 mW (100 % coupling).

Step 93 **quench_1033.sw.PD1** — Mount Intensity monitor (PDA36A2) at site 89, (603.0, 618.8) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Expect ≈ 1.5 mW on the detector.

Step 94 **spam_674.dp.HWP1** — Mount Input λ/2 (RSP05, WPH05M-780) at site 90, (100.0, 778.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Set the fast axis to 10.0° from horizontal (H).

Step 95 **spam_674.dp.PBS1** — Mount Polarizing beamsplitter cube (KM100CP, PBS052) at site 91, (125.0, 778.8) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Outputs: spam_674.dp.L1.t → 0.0° (20 mW); spam_674.dp.L1.t.+1.0.r → 89.1° (3.2 mW); spam_674.dp.L1.t.+1.+1.r → 90.0° (12.8 mW)

Step 96 **spam_674.dp.AOM1** — Mount 80 MHz AOM (Isomet 1205C-2) at site 92, (175.0, 778.8) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Drive at 92.61 MHz; use the +1 order (≈ 80 % efficiency).

Step 97 **spam_674.dp.LENS1** — Mount Cat-eye lens f=75 mm (AC127-075-B, LMR1) at site 93, (250.0, 779.9) mm; orientation 0.9°; keyed by pins, one 8-32 screw from below. f = 75 mm, 75.0 mm after the previous element on spam_674.dp.L1.t.+1.

Step 98 **spam_674.dp.BD2** — Mount 0th-order razor block (LB1) at site 94, (275.0, 779.2) mm; orientation 0.9°; keyed by pins, one 8-32 screw from below. Blocks: spam_674.dp.L1.t.0 (4 mW, 674 nm)

Step 99 **spam_674.dp.QWP1** — Mount λ/4 at 45° (RSP05, WPQ05M-780) at site 95, (300.0, 780.7) mm; orientation 0.9°; keyed by pins, one 8-32 screw from below. Set the fast axis to 45.0° from horizontal (H).

Step 100 **spam_674.dp.M1** — Mount Cat-eye retro mirror (BB05-E03, KM100) at site 96, (325.0, 781.0) mm; orientation 180.9°; keyed by pins, one 8-32 screw from below. Steer the beam to: spam_674.dp.L1.t.+1 → 180.9° (16 mW)

Step 101 **spam_674.dp.BD3** — Mount Return 0th-order ghost block (LB1) at site 97, (126.4, 823.1) mm; orientation 89.1°; keyed by pins, one 8-32 screw from below. Blocks: spam_674.dp.L1.t.+1.0.r (3.2 mW, 674 nm)

Step 102 **spam_674.dp.M2** — Mount Output fold mirror (BB05-E03, KM100) at site 98, (125.0, 853.8) mm; orientation 315.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: spam_674.dp.L1.t.+1.+1.r → 360.0° (12.8 mW)

Step 103 **spam_674.dp.FC1** — Mount Output fiber coupler (PAF2-A4A) at site 99, (340.0, 853.8) mm; orientation 360.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 4.6 mm, fiber MFD 4.58 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 12 mW (94 % coupling).

Step 104 **spam_674.dist.S1** — Mount 5% pickoff (BSF05-A, KM100) at site 100, (436.0, 865.0) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Outputs: spam_674.dist.L1.t → 0.0° (38 mW); spam_674.dist.L1.r → 270.0° (2 mW)

Step 105 **spam_674.dist.HWP1** — Mount lambda/2 tap 1 (RSP05, WPH05M) at site 101, (476.0, 865.0) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Set the fast axis to 15.0° from horizontal (H).

Step 106 **spam_674.dist.PBS1** — Mount Tap 1 PBS (KM100CP, PBS101) at site 102, (506.0, 865.0) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Outputs: spam_674.dist.L1.t.t → 0.0° (28.5 mW); spam_674.dist.L1.t.r → 270.0° (9.5 mW)

Step 107 **spam_674.dist.HWP2** — Mount lambda/2 tap 2 (RSP05, WPH05M) at site 103, (586.0, 865.0) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Set the fast axis to 17.6° from horizontal (H).

Step 108 **spam_674.dist.PBS2** — Mount Tap 2 PBS (KM100CP, PBS101) at site 104, (616.0, 865.0) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Outputs: spam_674.dist.L1.t.t.t → 0.0° (19 mW); spam_674.dist.L1.t.t.r → 270.0° (9.5 mW)

Step 109 **spam_674.dist.HWP3** — Mount lambda/2 tap 3 (RSP05, WPH05M) at site 105, (696.0, 865.0) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Set the fast axis to 22.5° from horizontal (H).

Step 110 **spam_674.dist.PBS3** — Mount Tap 3 PBS (KM100CP, PBS101) at site 106, (726.0, 865.0) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Outputs: spam_674.dist.L1.t.t.t.t → 0.0° (9.5 mW); spam_674.dist.L1.t.t.t.r → 270.0° (9.5 mW)

Step 111 **spam_674.dist.MZa** — Mount Last output steering 1 (BB05-E02, KM100) at site 107, (816.0, 865.0) mm; orientation 225.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: spam_674.dist.L1.t.t.t.t → 270.0° (9.5 mW)

Step 112 **spam_674.dist.MZb** — Mount Last output steering 2 (BB05-E02, KM100) at site 108, (816.0, 820.0) mm; orientation 45.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: spam_674.dist.L1.t.t.t.t → 0.0° (9.5 mW)

Step 113 **spam_674.dist.FC4** — Mount Output fiber 4 (PAF2-A4A) at site 109, (861.0, 820.0) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 4.6 mm, fiber MFD 4.59 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 8.76 mW (92 % coupling).

Step 114 **spam_674.dist.M3a** — Mount Tap 3 steering 1 (BB05-E02, KM100) at site 110, (726.0, 820.0) mm; orientation 45.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: spam_674.dist.L1.t.t.t.r → 0.0° (9.5 mW)

Step 115 **spam_674.dist.M3b** — Mount Tap 3 steering 2 (BB05-E02, KM100) at site 111, (766.0, 820.0) mm; orientation 225.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: spam_674.dist.L1.t.t.t.r → 270.0° (9.5 mW)

Step 116 **spam_674.dist.FC3** — Mount Output fiber 3 (PAF2-A4A) at site 112, (766.0, 770.0) mm; orientation 270.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 4.6 mm, fiber MFD 4.59 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 8.88 mW (93 % coupling).

Step 117 **spam_674.dist.M2a** — Mount Tap 2 steering 1 (BB05-E02, KM100) at site 113, (616.0, 820.0) mm; orientation 45.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: spam_674.dist.L1.t.t.r → 0.0° (9.5 mW)

Step 118 **spam_674.dist.M2b** — Mount Tap 2 steering 2 (BB05-E02, KM100) at site 114, (656.0, 820.0) mm; orientation 225.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: spam_674.dist.L1.t.t.r → 270.0° (9.5 mW)

Step 119 **spam_674.dist.FC2** — Mount Output fiber 2 (PAF2-A4A) at site 115, (656.0, 770.0) mm; orientation 270.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 4.6 mm, fiber MFD 4.59 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 9.13 mW (96 % coupling).

Step 120 **spam_674.dist.M1a** — Mount Tap 1 steering 1 (BB05-E02, KM100) at site 116, (506.0, 820.0) mm; orientation 45.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: spam_674.dist.L1.t.r → 0.0° (9.5 mW)

Step 121 **spam_674.dist.M1b** — Mount Tap 1 steering 2 (BB05-E02, KM100) at site 117, (546.0, 820.0) mm; orientation 225.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: spam_674.dist.L1.t.r → 270.0° (9.5 mW)

Step 122 **spam_674.dist.FC1** — Mount Output fiber 1 (PAF2-A4A) at site 118, (546.0, 770.0) mm; orientation 270.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 4.6 mm, fiber MFD 4.59 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 9.33 mW (98 % coupling).

Step 123 **spam_674.dist.PD1** — Mount Power monitor (PDA36A2) at site 119, (436.0, 820.0) mm; orientation 270.0°; keyed by pins, one 8-32 screw from below. Expect ≈ 2 mW on the detector.

Step 124 **cooling_422.dp.BD1** — Mount Dump for rejected V light (LB1) at site 120, (125.0, 33.8) mm; orientation 270.0°; keyed by pins, one 8-32 screw from below. Blocks: (nothing in this state)

Step 125 **spam_674.dp.BD1** — Mount Dump for rejected V light (LB1) at site 121, (125.0, 753.8) mm; orientation 270.0°; keyed by pins, one 8-32 screw from below. Blocks: (nothing in this state)

Alignment procedure

Align in beam order with an iris and an IR card; adjust one knob at a time and keep the best reading.

Order: cooling_422.dp.HWP1 → cooling_422.dp.AOM1 → cooling_422.dp.LENS1 → cooling_422.dp.QWP1 → cooling_422.dp.M1 → cooling_422.dp.M2 → cooling_422.dp.FC1 → cooling_422.dist.S1 → cooling_422.dist.HWP1 → cooling_422.dist.HWP2 → cooling_422.dist.MZa → cooling_422.dist.MZb → cooling_422.dist.FC3 → cooling_422.dist.M2a → cooling_422.dist.M2b → cooling_422.dist.FC2 → cooling_422.dist.M1a → cooling_422.dist.M1b → cooling_422.dist.FC1 → pi_405.fc.LENS1 → pi_405.fc.LENS2 → pi_405.fc.M1 → pi_405.fc.M2 → pi_405.fc.FC1 → raman_422.arm1.S1 → raman_422.arm1.LENS1 → raman_422.arm1.AOM1 → raman_422.arm1.LENS2 → raman_422.arm1.M1 → raman_422.arm1.M2 → raman_422.arm1.FC1 → raman_422.arm2.S1 → raman_422.arm2.LENS1 → raman_422.arm2.AOM1 → raman_422.arm2.LENS2 → raman_422.arm2.M1 → raman_422.arm2.M2 → raman_422.arm2.FC1 → pi_461.sas.HWP1 → pi_461.sas.FC1 → pi_461.sas.S1 → pi_461.sas.M2 → pi_461.sas.M3 → pi_461.sas.HWP2 → pi_461.fc.LENS1 → pi_461.fc.LENS2 → pi_461.fc.M1 → pi_461.fc.M2 → pi_461.fc.FC1 → repump_1092.sw.S1 → repump_1092.sw.LENS1 → repump_1092.sw.AOM1 → repump_1092.sw.LENS2 → repump_1092.sw.M1 → repump_1092.sw.M2 → repump_1092.sw.FC1 → quench_1033.sw.S1 → quench_1033.sw.LENS1 → quench_1033.sw.AOM1 → quench_1033.sw.LENS2 → quench_1033.sw.M1 → quench_1033.sw.M2 → quench_1033.sw.FC1 → spam_674.dp.HWP1 → spam_674.dp.AOM1 → spam_674.dp.LENS1 → spam_674.dp.QWP1 → spam_674.dp.M1 → spam_674.dp.M2 → spam_674.dp.FC1 → spam_674.dist.S1 → spam_674.dist.HWP1 → spam_674.dist.HWP2 → spam_674.dist.HWP3 → spam_674.dist.MZa → spam_674.dist.MZb → spam_674.dist.FC4 → spam_674.dist.M3a → spam_674.dist.M3b → spam_674.dist.FC3 → spam_674.dist.M2a → spam_674.dist.M2b → spam_674.dist.FC2 → spam_674.dist.M1a → spam_674.dist.M1b → spam_674.dist.FC1

No tuner run attached: use the design values listed in the assembly steps.

Expected readings

- expect ≈ 12.5 mW at cooling_422.dp.FC1, coupling 98 %
- expect ≈ 2 mW at cooling_422.dist.PD1
- expect ≈ 12.6 mW at cooling_422.dist.FC1, coupling 99 %
- expect ≈ 12.5 mW at cooling_422.dist.FC2, coupling 98 %
- expect ≈ 12.4 mW at cooling_422.dist.FC3, coupling 98 %
- expect ≈ 9.9 mW at pi_405.fc.FC1, coupling 99 %
- expect ≈ 1.5 mW at raman_422.arm1.PD1
- expect ≈ 22.8 mW at raman_422.arm1.FC1, coupling 100 %
- expect ≈ 1.5 mW at raman_422.arm2.PD1
- expect ≈ 22.8 mW at raman_422.arm2.FC1, coupling 100 %
- expect ≈ 18 mW at pi_461.sas.FC1, coupling 100 %
- expect ≈ 0.12 mW at pi_461.sas.PD1
- expect ≈ 9.87 mW at pi_461.fc.FC1, coupling 99 %
- expect ≈ 1.5 mW at repump_1092.sw.PD1
- expect ≈ 22.7 mW at repump_1092.sw.FC1, coupling 100 %
- expect ≈ 1.5 mW at quench_1033.sw.PD1
- expect ≈ 22.7 mW at quench_1033.sw.FC1, coupling 100 %

- expect ≈ 12 mW at spam_674.dp.FC1, coupling 94 %
- expect ≈ 2 mW at spam_674.dist.PD1
- expect ≈ 9.33 mW at spam_674.dist.FC1, coupling 98 %
- expect ≈ 9.13 mW at spam_674.dist.FC2, coupling 96 %
- expect ≈ 8.88 mW at spam_674.dist.FC3, coupling 93 %
- expect ≈ 8.76 mW at spam_674.dist.FC4, coupling 92 %

Laser safety

Wavelengths: 405 nm, 421.7 nm, 460.9 nm, 674 nm, 1033 nm, 1091.8 nm. **Maximum source power on the table:** 40 mW.

Wavelength	Band	Eyewear (generic guidance)
405 nm	visible	OD ≥ 3 at 405 nm (generic; confirm with your laser safety officer)
422 nm	visible	OD ≥ 3 at 422 nm (generic; confirm with your laser safety officer)
461 nm	visible	OD ≥ 3 at 461 nm (generic; confirm with your laser safety officer)
674 nm	visible	OD ≥ 3 at 674 nm (generic; confirm with your laser safety officer)
1033 nm	near-IR (invisible!)	OD ≥ 3 at 1033 nm (generic; confirm with your laser safety officer)
1092 nm	near-IR (invisible!)	OD ≥ 3 at 1092 nm (generic; confirm with your laser safety officer)

Beam dumps

Dump	Blocks
cooling_422.dp.BD2	cooling_422.dp.L1.t.0 (4 mW, 421.7 nm)
cooling_422.dp.BD3	cooling_422.dp.L1.t.+1.0.r (3.2 mW, 421.7 nm)
raman_422.arm1.BD1	raman_422.arm1.L1.t.0 (5.7 mW, 421.7 nm)
raman_422.arm2.BD1	raman_422.arm2.L1.t.0 (5.7 mW, 421.7 nm)
pi_461.sas.BD1	pi_461.sas.L1.r.t.r.r (1.08 mW, 460.9 nm)
repump_1092.sw.BD1	repump_1092.sw.L1.t.0 (5.7 mW, 1091.8 nm)
quench_1033.sw.BD1	quench_1033.sw.L1.t.0 (5.7 mW, 1033 nm)
spam_674.dp.BD2	spam_674.dp.L1.t.0 (4 mW, 674 nm)
spam_674.dp.BD3	spam_674.dp.L1.t.+1.0.r (3.2 mW, 674 nm)
cooling_422.dp.BD1	(nothing in the default state)
spam_674.dp.BD1	(nothing in the default state)

Verify all beam dumps before full power. Align at the lowest usable power, never look along the beam, remove watches and rings, keep the beam below eye level, and enclose invisible (IR) beams.

Credits

- Layout concept and “paint by numbers” baseplates: PyOpticL (code-to-CAD optics).
- Designed by Gemini `gemini-4-argon` agents, verified by the Beamline tracer.
- Prices are list-price estimates; part numbers should be checked against current vendor catalogs.